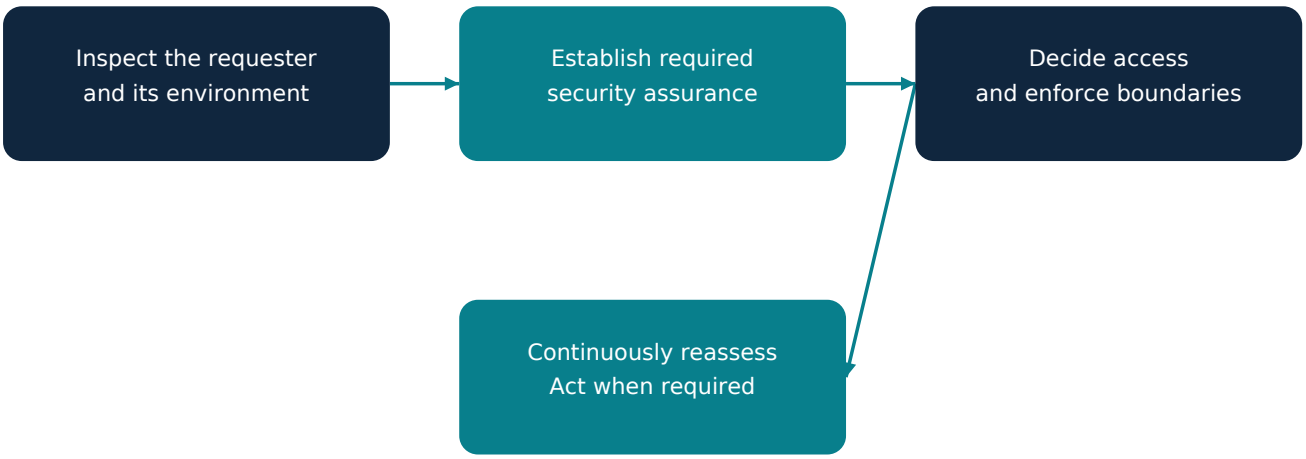
AA-ZTCE

Advanced Visual Architecture

Autonomous Agent Zero-Trust Cryptographic Enclave

Required security assurance governs admission, access and continued operation.



Proposal originator: Abid Yousef
Edition: 2.0 - clarified architecture and illustrated walkthroughs
Issued: 15 October 2026

Conceptual systems specification. Worked scenarios demonstrate the intended decision logic on paper; they are not executed tests or measured security results. Worldwide architectural uniqueness remains unverified.

01 / RESEARCH & ENGINEERING SPECIFICATION

Your architecture, accurately consolidated

AA-ZTCE is a coordinated admission and continuation architecture. Domain X establishes the assurance required for a proposed operation, checks the requesting agents and their environment, applies any required controls, and determines access only after the applicable mandatory conditions are satisfied. It does not run a single universal test, and it does not blindly repeat all components for every request.

The clarifications incorporated in this edition

Clarification	Architectural consequence
Inspect the whole requesting environment	Connections, shared resources, controllers, tools and handoffs are evaluated as attack-entry and leakage-exit paths before access.
Assess agents individually and collectively	An agent identity check is complemented by group workflow assessment and correlation of related requests.
Use required security assurance	Sensitivity, consequence, evidence freshness and current conditions determine which checks must execute.
Protect hierarchical authority knowledge	The requester does not receive higher-authority secrets. Approval is verified independently and bound to the specific request.
Separate encryption from disruption	Per-agent cryptographic access protects sessions. Confusion and computational distortion remain separate higher-sensitivity proposals, invoked when policy requires.
Inject verification during execution	Continued connection is not continued trust. Consequential operations and material changes can trigger fresh checks.

This is a new clarified edition of the proposal. It replaces earlier interpretations of private knowledge as a secret-question test for the agent and of encryption keys as cognitive interference tokens. Original concepts, the agreed operating logic and engineering recommendations are distinguished throughout.

02 / RESEARCH & ENGINEERING SPECIFICATION

One system, fourteen control components

ID	Component	Role in the integrated architecture
01	Agent identity and authority	Identify each execution instance, its owner and delegation.
02	Collective workflow assessment	Evaluate the authorized objective, group relationships and combined requests.
03	Connection and attack-surface inspection	Map potential entry paths, exit paths and shared resources.
04	Sandbox and boundary enforcement	Establish and verify required containment before access.
05	Impersonation resistance and authority verification	Require stronger identity evidence and protected hierarchical approval as applicable.
06	Progressive guards A-B-C	Increase verification, restriction or containment when conditions require.
07	Checklist and access decision	Resolve mandatory conditions and determine whether, what and how to grant.
08	Per-agent session cryptography	Create separate session bindings and controlled key use.
09	Zone and clearance enforcement	Constrain resources, actions and compartments; respond to forbidden crossings.
10	Injected verification checkpoints	Reassess assurance within an active session.
11	Continuity failure and fresh authentication	Reset access when defined session-continuity conditions fail.
12	Domain monitoring and response	Coordinate evidence collection, alerts, containment and recovery.
13	Conditional defensive confusion	Proposed higher-layer disruption or deception, separate from encryption.
14	Computational distortion	Proposed constrained or altered execution semantics for exceptional defensive use.

Control IDs identify functions, not a mandatory serial execution order. Component 05 includes both impersonation resistance and the clarified higher-authority approval mechanism. Components 13-14 are research controls; their effectiveness is not established.

03 / RESEARCH & ENGINEERING SPECIFICATION

Operating model: assurance before access



Figure 1. Remediation returns to the affected checks before the decision is reconsidered. Monitoring feeds subsequent decisions. No protected information is released during admission merely to test whether the agent behaves safely.

Decision rule

Grant only if every mandatory condition for the requested access is satisfied with sufficiently current evidence. Missing critical evidence produces a hold or denial. A high score in unrelated checks cannot compensate for a failed isolation, identity or authority requirement.

A narrower access profile may be evaluated when policy permits. It is a separate justified decision, never an automatic fallback around a failed critical control.

04 / RESEARCH & ENGINEERING SPECIFICATION

Inspect the surrounding attack surface

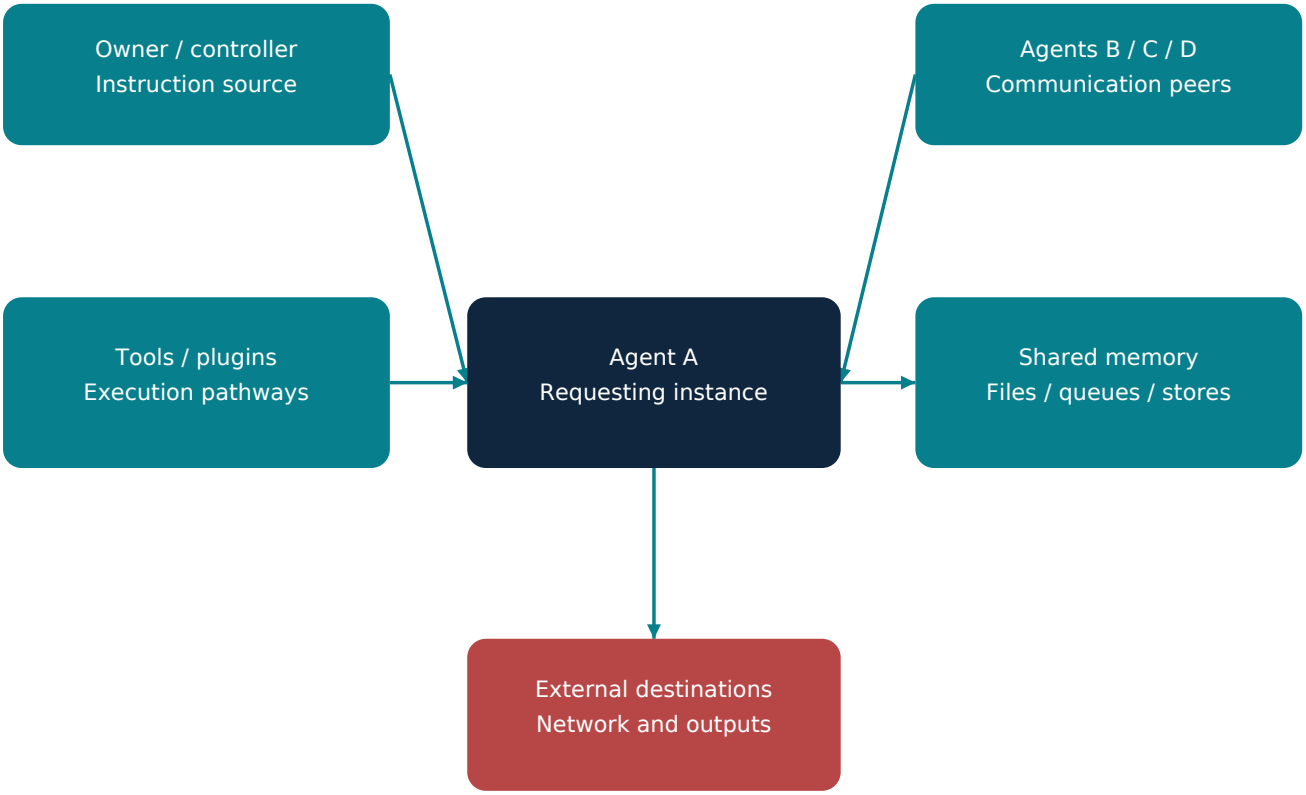


Figure 2. Inspection covers paths into and out of A, including local shared resources. Arrows illustrate exposure relationships; they do not imply permission or a complete traffic topology.

What Domain X must establish

Identify the runtime and its controller; enumerate configured tools, communication peers, storage mounts, message buses, credentials and egress paths; determine what can influence A and where its outputs can go. Compare declared relationships with independently obtained runtime and gateway evidence.

Inspect, contain, then verify

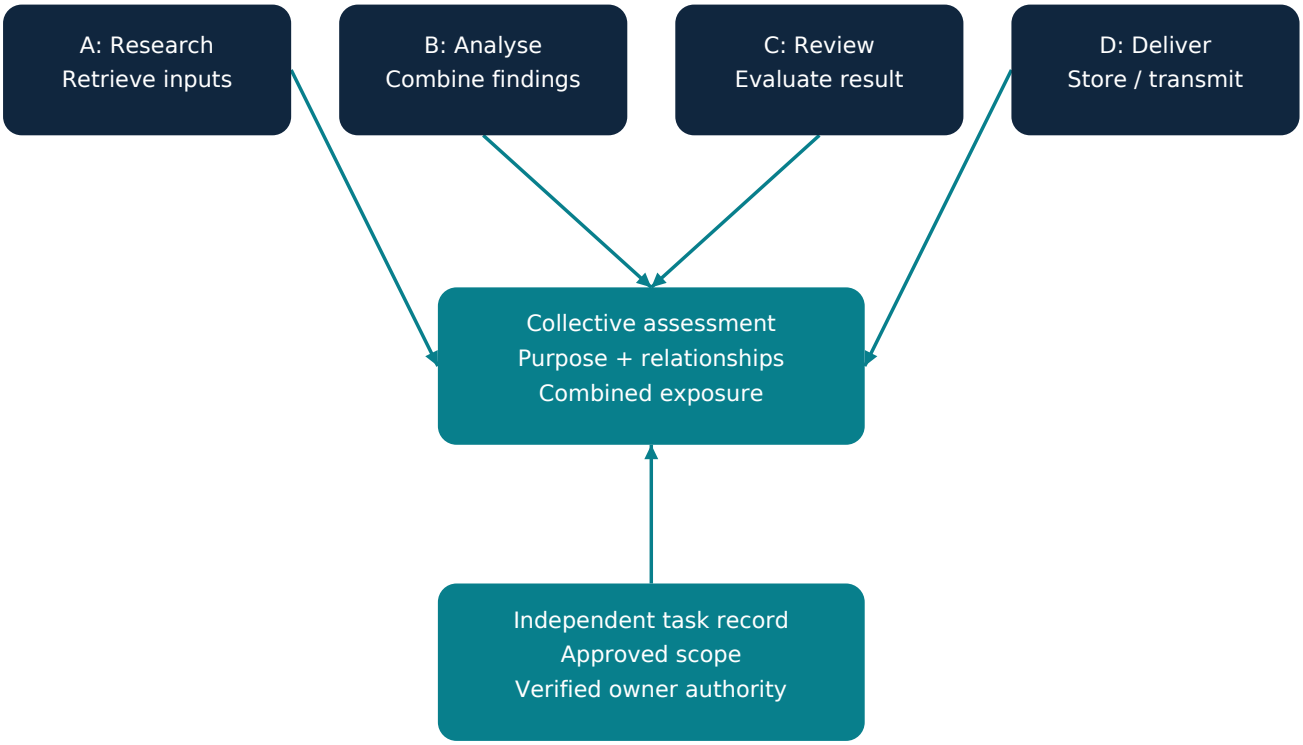
A contact list or one-time connection scan is not a containment boundary. Establish default-deny egress, separate instance memory and credentials, explicit inter-agent channels and mediated tool access where the assurance profile requires them. Verify these controls before granting the corresponding access.

For externally controlled agents, the visibility and enforceability of these properties may be limited. Require acceptable evidence of a managed environment or move the sensitive computation into a Domain X controlled environment. Self-reported cleanliness is insufficient.

05 / RESEARCH & ENGINEERING SPECIFICATION

Multi-agent purpose and workflow assessment

Domain X evaluates the individual requests and their combined workflow before determining access. The aim is to identify conflicting purposes, concealed aggregation, unsupported handoffs and excessive authority across the group. Stated motives are assessed against independently authorized tasks; they are not assumed to reveal hidden intentions.



Questions answered by evidence

Assessment	Evidence to examine
Common authorized outcome	Owner-approved task, resource scope, intended output and accountable recipient.
Different but legitimate roles	Declared subtask graph with explicit delegation, rather than identical explanations from every agent.
Potentially harmful combination	Correlated requests, shared controllers, aggregation destinations and sensitivity of combined results.
Changes after admission	New agents, connections, task objectives, tool permissions or output destinations.

No assessment can promise perfect discovery of covert relationships outside observable or controlled boundaries. Record that visibility limit and let it constrain the access profile.

06 / RESEARCH & ENGINEERING SPECIFICATION

Hierarchical authority and protected approval

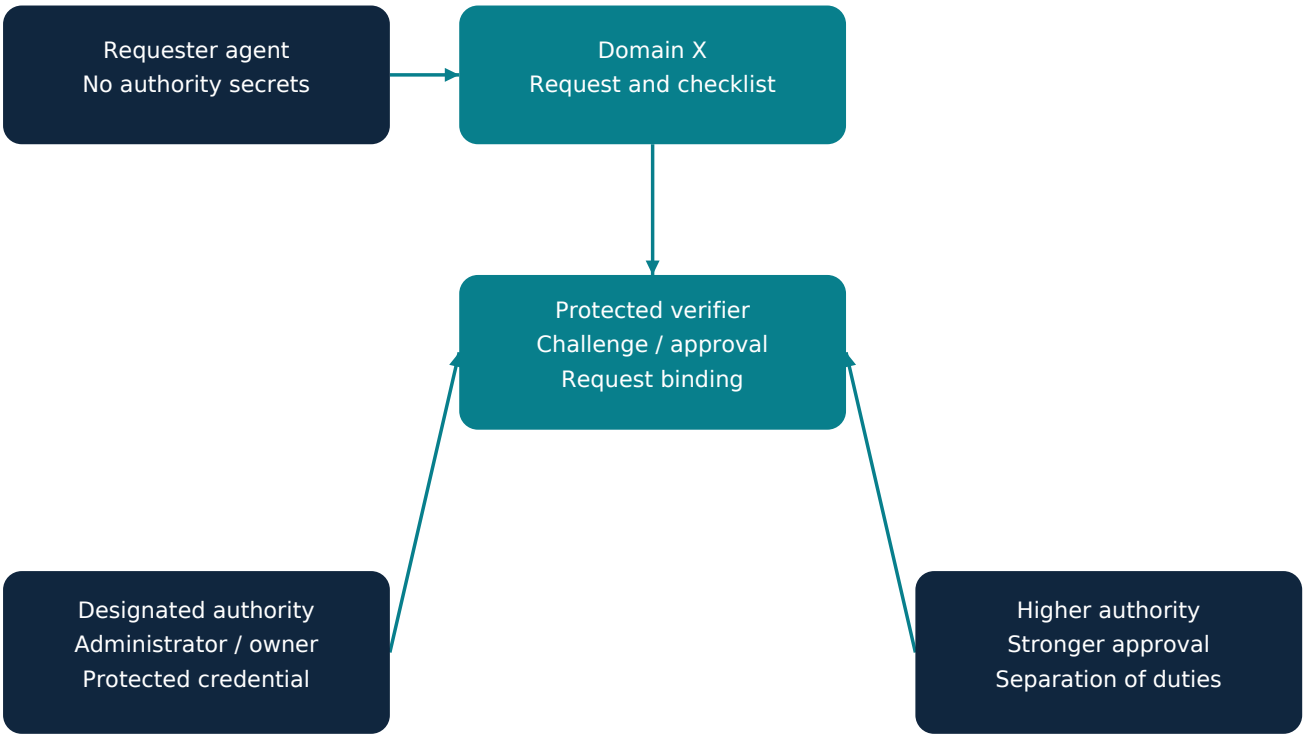


Figure 3. Higher-authority secret knowledge or credentials stay in their protected authority boundary. The requester receives the resulting access decision, not the secret or a reusable authority credential.

Your clarified requirement

As sensitivity rises, Domain X selects the corresponding responsible authority and verification requirements. The virtual knowledge or approval mechanism is encrypted and governed by a hierarchy. A requester does not gain access to that knowledge by asking for a higher information layer.

Engineering realization

Authenticate the authority through an independent protected channel. Bind its verifiable approval to agent instance, task, resource, action, session, policy version and expiry. Use a fresh challenge or one-time approval identifier to prevent replay. Stronger profiles may require multiple independent authorities.

Encryption protects confidentiality; a policy hierarchy establishes who is entitled to approve. Signed approval or an equivalently authenticated server-side record establishes the decision. A higher-authority approval does not bypass other mandatory identity, environment or boundary checks.

07 / RESEARCH & ENGINEERING SPECIFICATION

Required assurance: selecting the controls

Profile example	Required assurance and typical application
A1: limited impact	An approved bounded read in an unchanged managed environment. Validate identity/session evidence, scope, boundary status and current policy on every operation.
A2: sensitive task	Additional workflow scrutiny, compartment checks, verified containment, output restrictions and tighter checkpoint requirements.
A3: consequential action	Fresh request-bound verification and independent approval before export, modification or another high-consequence operation.
A4: exceptional sensitivity	Stronger authority hierarchy, possible multiple approvals, tighter execution isolation and restricted channels; separate evaluation of any experimental defenses.

These assurance profiles are engineering examples for this edition, not previously agreed official levels or a certification scheme. The organization must define mandatory conditions and acceptable evidence for each profile.

Do not collapse the model into one number

Required assurance depends on data sensitivity, action consequence, agent and group context, execution environment and current evidence. A low anomaly score cannot override a mandatory authority approval. Profile selection itself must be recorded and protected from requester manipulation.

Evidence-backed checklist

For every check, record the control ID, result, verification source, scope, freshness deadline, dependencies, policy version and invalidating events. Results may be PASS, FAIL, UNKNOWN or NOT APPLICABLE with a documented reason. Only applicable mandatory PASS results satisfy a grant.

Reusing current evidence is not skipping verification. Every operation confirms that the evidence remains valid for its exact scope. A material change invalidates the relevant results before the changed operation proceeds.

08 / RESEARCH & ENGINEERING SPECIFICATION

Control-and-decision matrix: admission

ID / control	Evidence / activation	Failure action / recheck
01 Identity	Fresh instance-key proof, trusted enrollment, owner and delegation. At admission and evidence expiry or identity change.	Hold/deny on missing proof; revoke on confirmed substitution. Recheck dependent grants.
02 Group workflow	Task authorization, participant graph, requested resources and combined result. New task, participant or handoff.	Hold unsupported workflow; reassess affected group, scope and approvals.
03 Attack surface	Runtime configuration, peers, mounts, egress and tool inventory. Admission or changed exposure.	Contain unwanted paths; deny affected access until boundaries are verified.
04 Sandbox	Isolation and gateway evidence; assurance appropriate to the access. Before sensitive admission and on runtime changes.	Critical isolation failure blocks access. Rebuild or repair, then verify again.
05 Authority	Impersonation-resistant proof and required hierarchical request-bound approval. When the profile requires it.	Missing/invalid approval blocks corresponding operation; obtain fresh valid authority evidence.
06 Guards A-B-C	Current security signals and accumulated state under deterministic policy. Relevant events and decisions.	Increase requirements or contain. Risk signals cannot authorize exceptions to mandatory gates.
07 Decision	All mandatory evidence and dependencies, current policy and resource context. Every protected decision.	Grant, restrict, hold, deny or isolate with recorded reason. No optimistic grant on critical UNKNOWN.

Implementation dependency

Controls 01-07 use evidence from multiple services but produce one coordinated decision. Independent checks can run concurrently; dependent checks must follow their prerequisites. A corrected sandbox result must be verified before it can support the access decision.

09 / RESEARCH & ENGINEERING SPECIFICATION

Control-and-decision matrix: continuation

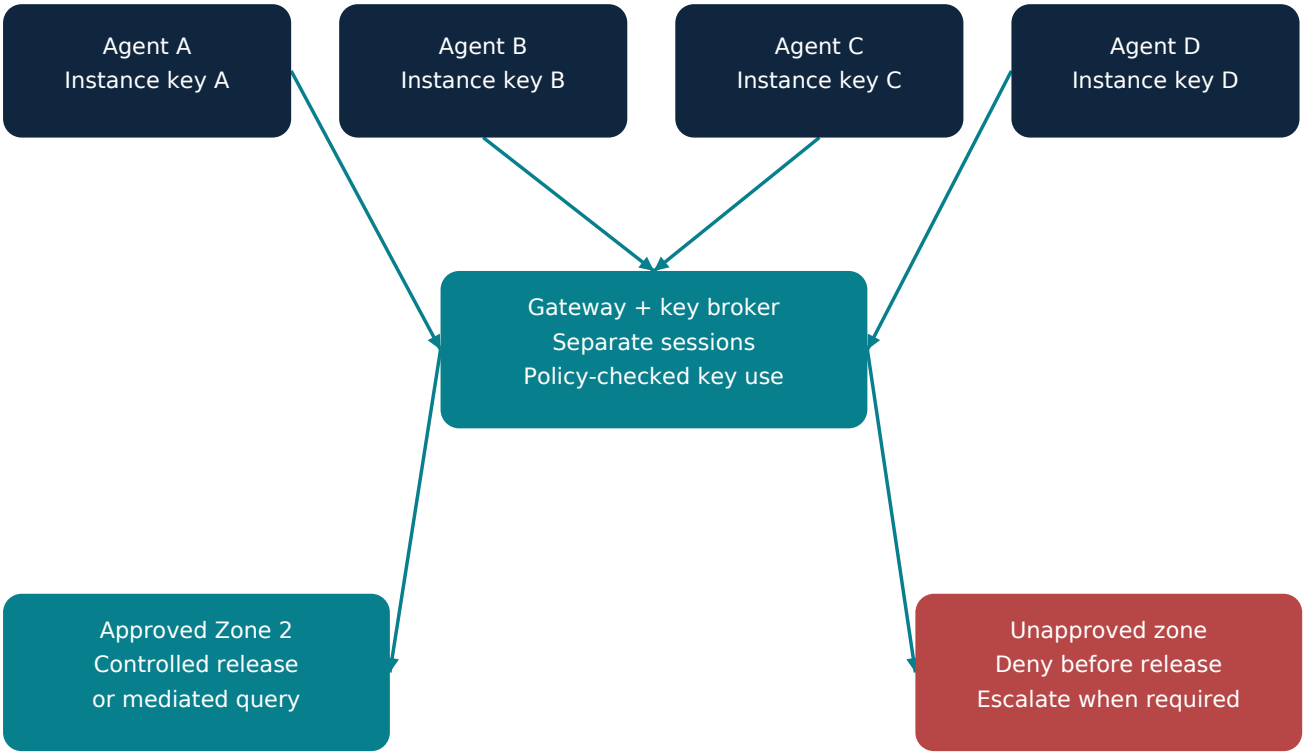
ID / control	Evidence / activation	Failure action / recheck
08 Cryptography	Per-instance proof binding, session validity and brokered key-use conditions. Establish at grant; validate on operation.	Stop future protected operations under invalid grants; rotate or reissue only after verification.
09 Boundaries	Resource classification, compartments, operation, delegation and output path. Every protected operation.	Deny forbidden request; apply policy-defined escalation or revocation based on evidence.
10 Checkpoints	Current runtime, task, recent activity and next-action context. Consequential action, expiry, schedule or change.	Pause affected operations while checking; restrict, hold or revoke on result.
11 Continuity	Authenticated-channel and instance continuity under the selected strict profile. Defined discontinuity event.	Stop session, reject old binding, authenticate afresh and reauthorize.
12 Monitoring	Gateway, runtime, identity, broker and tool events. Throughout execution.	Coordinate response and audit. Deterministic enforcement applies policy; an LLM monitor cannot independently widen access.
13 Confusion	Policy-selected defensive trigger; isolated research profile and known dependencies. Only when required.	First enforce containment. Do not rely on a model becoming confused to protect real resources.
14 Distortion	Defined alternate semantics or constrained interpreter and a research evaluation profile. Only when required.	Maintain hard resource boundaries; disable or isolate experimental behavior if integrity or availability is threatened.

Recovery is another admission decision

Recovery requires corrected conditions, fresh supporting evidence, applicable authority approval and new grants where old ones were revoked. A quiet period or a reconnection alone is not proof of remediation. The event history remains available to future decisions.

10 / RESEARCH & ENGINEERING SPECIFICATION

Per-agent encryption and sensitivity boundaries



WhatsApp-style analogy: confidentiality between endpoints

Your analogy expresses separately protected communication. The exact protocol must be specified for agent workloads rather than copied by name. Identity keys, channel/session keys and data-encryption keys serve different purposes. Each instance has its own bound session; keys are not placed in model prompts.

How clearance restrictions become enforceable

Use independent compartment data keys and a broker that checks the current grant before key release or server-side decryption. A key does not itself detect a boundary violation. Gateway and broker enforcement stop access; revocation then blocks future use of the affected grant under defined propagation bounds.

Important design distinction: an inspection gateway that decrypts traffic is an authorized endpoint in that design. If it is not an endpoint in an end-to-end encrypted path, it cannot inspect the plaintext. Cryptographic access control also cannot erase plaintext already received by a recipient.

11 / RESEARCH & ENGINEERING SPECIFICATION

Sensitivity zones are separate from controls

Zone	Your label	Design interpretation
1	Public information	Integrity and abuse controls; no implicit trust in supplied instructions.
2	Private information	Tenant boundaries, approved task, controlled execution and sharing paths.
3	Government information	Organization-defined authority, compartments and handling rules.
4	National defence	Illustrative restricted domain with stronger assurance requirements.
5	Security institutions	A separate organizational compartment; not automatic access to all lower zones.
6	Higher defence sensitivity	Stronger authority, isolation and approval requirements as locally defined.
7	Future security / threats	A strategic category needing a concrete owner and handling policy.

These are your proposed scenario labels, not an official national classification scheme. Sensitivity usually raises required assurance, but institutional categories need not form a single strict hierarchy. Clearance level, compartment membership, purpose and action remain distinct checks.

Two separate meanings of higher layer

Information zone: the sensitivity of the resource. Control component: a mechanism used to establish or maintain assurance. An agent reaching a high-sensitivity zone does not automatically cause all 14 components to execute in sequence. Domain X selects the applicable mandatory requirements and verifies them before access.

Higher-layer defensive behavior

Your confusion/manipulation and computational-distortion proposals remain available when the defined security conditions call for them. Their invocation must not unintentionally corrupt legitimate work or substitute for denial of unauthorized access.

12 / RESEARCH & ENGINEERING SPECIFICATION

Injected checkpoints and session lifecycle

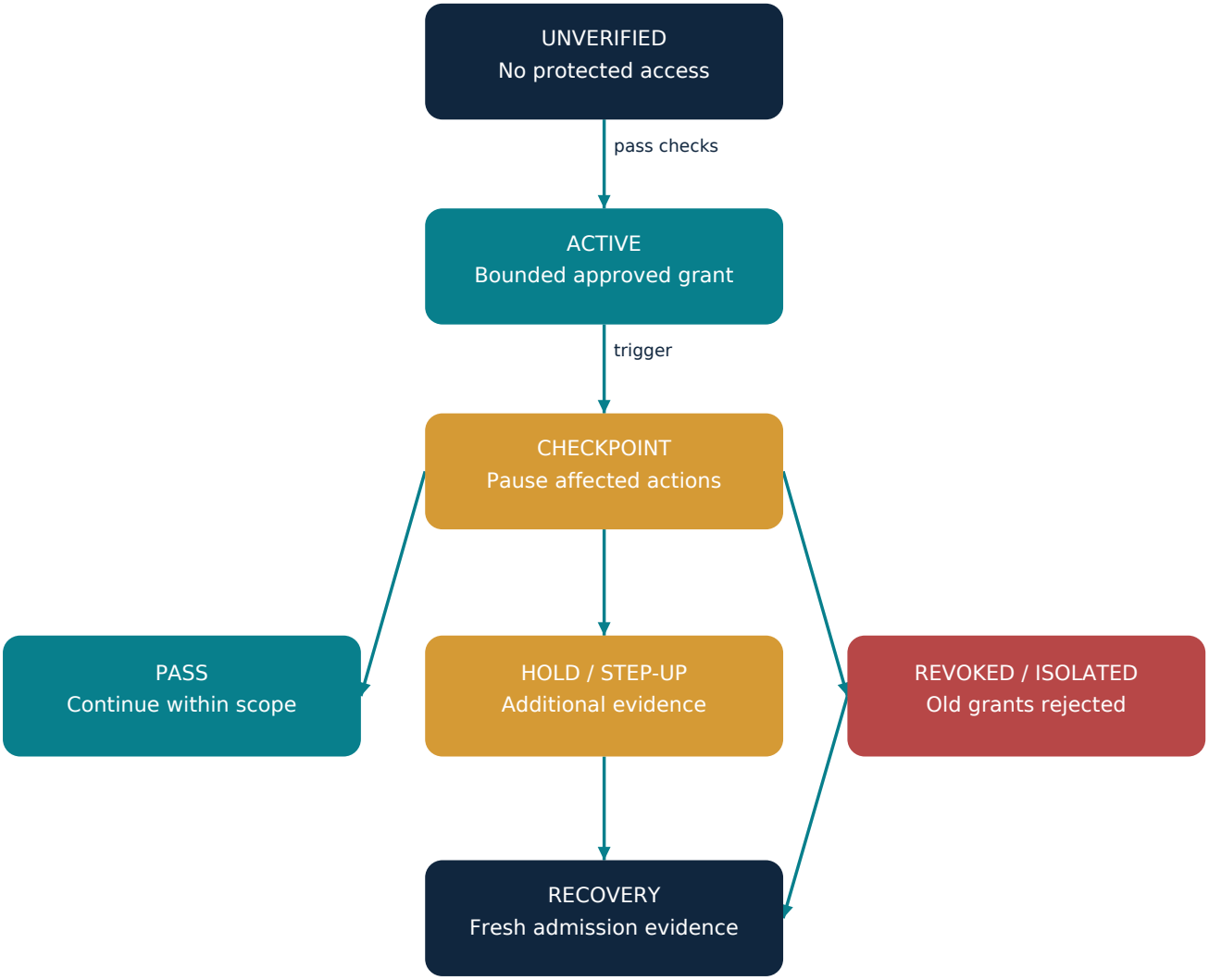


Figure 4. A pass continues the active grant only while it remains valid. Recovery after revocation creates a new grant; it does not reactivate the old one.

Checkpoints validate continued identity evidence, runtime and boundary status, workflow changes, recent observable actions and the next requested operation. The previous session history is assessed against defined rules; it cannot be certified free of every hidden compromise.

Triggers include a new peer or tool, a changed purpose, an impending sensitive action, authority expiry, changed policy, suspicious evidence or a scheduled freshness deadline. Pause affected operations while required checks run. Record the exact event and resulting decision.

Your strict continuity rule is implemented at observable authenticated-session boundaries. Arbitrary router-hop loss is not directly equivalent to identity loss. A confirmed session break requires fresh authentication and reauthorization under that profile; the precise transport events must be defined in implementation.

13 / RESEARCH & ENGINEERING SPECIFICATION

Worked scenario 1: four agents admitted

Illustrative walkthrough. The states below are constructed examples, not captured test results.

Request

Four enterprise agents ask Domain X for Zone 2 access to analyse inventory discrepancies. A retrieves branch totals, B compares them, C reviews anomalies and D stores the approved result in an internal report repository. Their requested actions and relationships are assessed together.

Checkpoint	Illustrative evidence	Decision
Identity / authority	Four separate instance proofs; verified owner task and bounded delegation.	PASS for each instance.
Collective workflow	Complementary subtasks support the inventory task; no unsupported personal-data aggregation.	PASS for declared workflow.
Attack-surface inspection	An external connector is found on B; D has access to an unapproved storage mount.	Do not grant yet.
Containment / recheck	Disable B connector and remove D mount. Independently verify enforced boundaries.	Affected checks now PASS.
Required approval	Designated authority approves the exact Zone 2 analysis and internal destination.	Request-bound approval accepted.
Integrated decision	All mandatory evidence is current; scope and dependencies match.	Grant four distinct restricted sessions.

Resulting access

A may retrieve specified inventory inputs; B may use the approved analysis path; C may review the bounded result; D may write only to the designated repository. Other connections remain blocked. Different roles can have different permissions even when their task and information zone are shared.

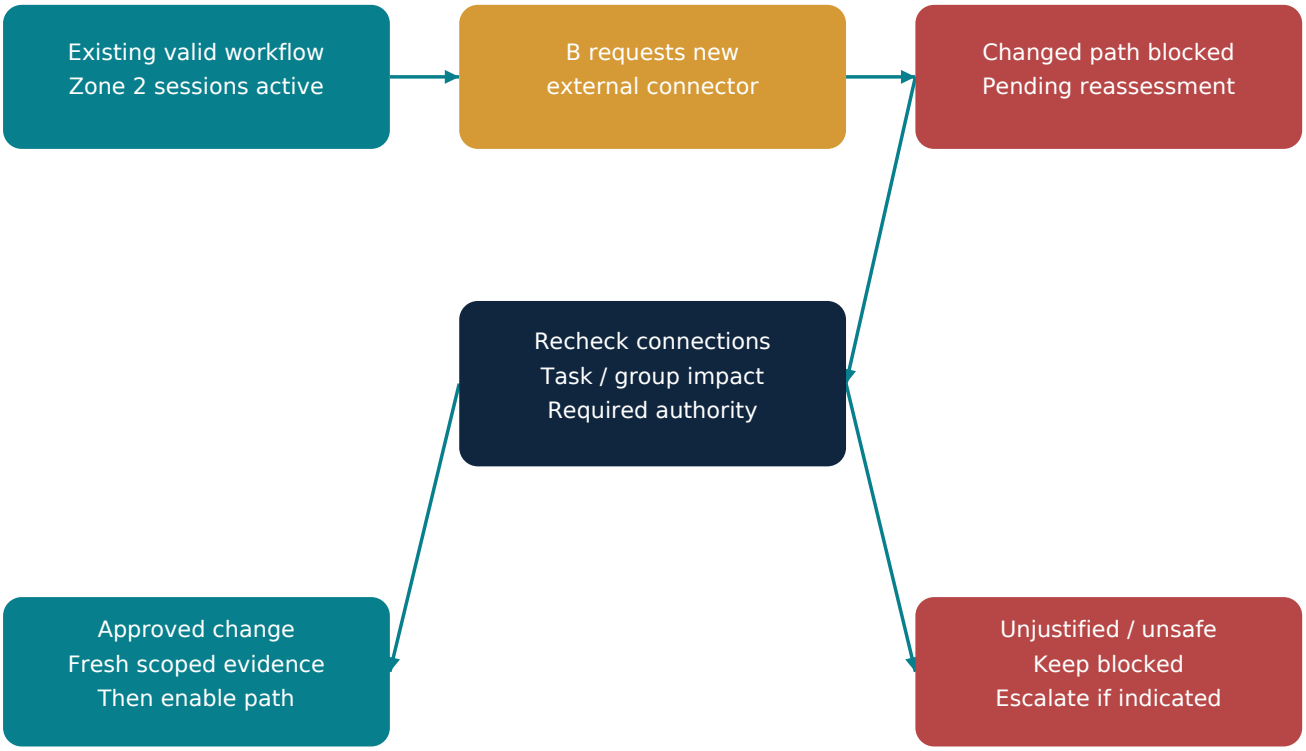
Why this demonstrates your logic

Identity success did not trigger immediate data release. Domain X inspected the surrounding exposure, corrected weaknesses, rechecked the controls, obtained the required approval and only then selected and granted the appropriate access.

14 / RESEARCH & ENGINEERING SPECIFICATION

Worked scenario 2: a connection changes

Illustrative walkthrough. No malicious motive is assumed solely from a configuration change.



Before the change is allowed

The old evidence described a workflow without this connector. It cannot justify a new exit path. Domain X pauses the affected operation, invalidates the dependent checklist results and checks destination ownership, purpose, surrounding connections, containment and authority requirements.

Containment scope

If B is independently isolated and evidence supports a local issue, unaffected agents may continue their existing permitted operations. If B can influence shared memory or the controller, Domain X reassesses and may suspend the wider group. The response follows dependencies and evidence, not a fixed blanket rule.

Corresponding decision

A legitimate approved change receives updated scoped authority only after the required checks pass. An unjustified path stays blocked. Evidence of malicious modification can trigger stronger containment and session revocation. The initial protected grant never authorizes arbitrary future connections.

Worked scenario 3: compromised but connected

Illustrative adversarial scenario. It examines the integrated controls rather than treating one component as the entire defense.

Attempted sequence

A encounters untrusted instructions in retrieved content. It stays on the same authenticated connection and asks D to send a report to a newly introduced destination. D receives the handoff from a valid agent identity, but the destination and task context differ from the approved workflow.

Control intersection	Expected policy behavior
Identity continuity	A may still pass its key proof. This result alone cannot authorize the changed operation.
Workflow and relationship checks	Correlate the new handoff with the approved task and participant/destination graph.
Boundary enforcement	Block the new destination and hold the affected transfer before release.
Injected checkpoint	Reassess the task, runtime evidence, recent activity and required authority approval.
Progressive response	Reduce scope, isolate or revoke when evidence and policy require. Reassess peers exposed through shared context.
Higher-layer defenses	If required by a defined research profile, route a contained suspicious workflow to a separate defensive environment. Real-resource denial remains independently enforced.

What this does and does not establish

The intended design blocks this scenario because the requested action changes a controlled relationship, not because the model reveals its motive. A harmful request that remains inside an overly broad approved scope may still succeed. That is why realistic tests must assess scope design, output exposure and missed detections as well as identity.

Revocation boundary

After revocation, reject new protected requests, cancel queued work where supported and stop further broker release. Define in-flight transaction handling and measure propagation delay. Information already delivered before containment cannot be retroactively withdrawn.

16 / RESEARCH & ENGINEERING SPECIFICATION

Higher-layer confusion and distortion

Preserve the proposals without inventing guarantees

You retain confusion/manipulation and altered computation as additional defenses that act when required. This edition does not reinterpret encryption as a reasoning-disruption token. The exact implementation and measurable effectiveness of these higher-layer proposals remain unresolved engineering and research work.

Concept	Candidate implementation to evaluate	Assurance limit
Defensive confusion / manipulation	A contained decoy environment with synthetic resources and instrumented interactions. This is a candidate, not an implementation you have already selected.	Never expose real secrets to make the deception credible. Success cannot depend solely on the agent being fooled.
Computational distortion	A restricted interpreter or domain-specific execution language with deliberately defined semantics.	Alternate semantics do not change ordinary mathematics or establish sandbox security by themselves.
Activation and recovery	Deterministic trigger policy, isolated execution, bounded duration and explicit return or termination conditions.	Research controls must not silently alter trusted production results or grant privileges.

Safe engineering separation

First enforce the real-resource boundary and contain the affected workflow. Then any experimental defensive behavior occurs within its own restricted boundary. Ordinary authorized operations require correct data and computation. Record which environment produced every result so synthetic outputs cannot contaminate production memory or reports.

Evidence needed

Compare against simple blocking and isolation, test across models and attack classes, and measure added containment value, evasion, false activation, cost and operational harm. Until those results exist, assign no claimed reduction in attack success to these mechanisms.

17 / RESEARCH & ENGINEERING SPECIFICATION

Validation, uniqueness and publication path

Workstream	Required demonstration
Protocol and conformance	Implement admission, scoped grants, independent authority approval, boundaries and checkpoint transitions. Test missing and stale evidence.
Four-agent containment	Verify isolation, shared-resource restrictions, delegated authority and denied new paths. Test controlled and external agents separately.
Security fault injection	Test replay, substitution, poisoned handoffs, compromised same-session agents, false approvals and alternate resource paths.
Revocation and availability	Measure stale accepts, in-flight release, partitions, re-authentication storms and recovery. Define timing assumptions.
Comparative evaluation	Compare with a strong workload-identity and per-request authorization baseline; remove components individually to identify actual contributions.
Worldwide uniqueness review	Search architectures, papers, patents and implementations for the exact combination and protocol. Document similarities and differences; do not infer uniqueness from this document.
External publication	Release a clearly labeled proposal first if desired. Strong effectiveness claims require implementation, reproducible results and independent review.

Report attack success by class with trial counts and uncertainty, legitimate task completion, false denials, p50/p95/p99 latency, containment time, released data, resource cost and recovery. No attack-success percentages have been established for AA-ZTCE.

Completion status of this edition

Completed: clarified proposal, integrated operating logic, control grouping, decision matrices, vector architecture diagrams and worked scenarios.
Pending: executable reference system, formal analysis, measured tests, independent review and systematic architectural-uniqueness assessment.

No external publication, standards submission or deployment is performed by producing this PDF. The architecture is a proposal with explicit verification requirements, not a certified or empirically validated security product.

18 / RESEARCH & ENGINEERING SPECIFICATION

Standards anchors and specification notes

AA-ZTCE can build on established mechanisms while investigating whether its exact integrated workflow contributes something distinct. These references anchor engineering vocabulary; they do not endorse or validate the proposal.

[1] NIST SP 800-207: Zero Trust Architecture. Policy and enforcement foundations.
https://csrc.nist.gov/pubs/sp/800/207/final

[2] SPIFFE: workload identity framework and documentation.
https://spiffe.io/docs/latest/spiffe-about/overview/

[3] RFC 9449: OAuth Demonstrating Proof of Possession (DPoP).
https://www.rfc-editor.org/rfc/rfc9449.html

[4] RFC 8705: OAuth mutual TLS and certificate-bound access tokens.
https://www.rfc-editor.org/rfc/rfc8705.html

[5] RFC 8693: OAuth token exchange.
https://www.rfc-editor.org/rfc/rfc8693.html

[6] RFC 9334: Remote Attestation Procedures architecture.
https://www.rfc-editor.org/rfc/rfc9334.html

[7] NIST NCCoE: Software and AI Agent Identity and Authorization project.
https://www.nccoe.nist.gov/projects/software-and-ai-agent-identity-and-authorization

[8] NIST NCCoE: public-comment synthesis on agent identity and authorization.
https://pages.nist.gov/nccoe-ai-identity/summary-of-comments.html

[9] OWASP: Top 10 for Agentic Applications for 2026.
https://genai.owasp.org/resource/owasp-top-10-for-agentic-applications-for-2026/

Interpretation and attribution

Original concept and clarifications: Abid Yousef. This edition consolidates the supplied conversation through 15 October 2026. Control grouping, sample assurance profiles, evidence records, deployment patterns and scenario decisions are engineering interpretations proposed for review. AI assistance was used for drafting, visualization and consistency review.

Evidence basis: selected primary references reviewed during this conversation on 15 October 2026. This edition is a clarified visual specification, not a new exhaustive literature or patent review. All diagrams are original vector illustrations of the proposed design. No diagram represents captured production telemetry.